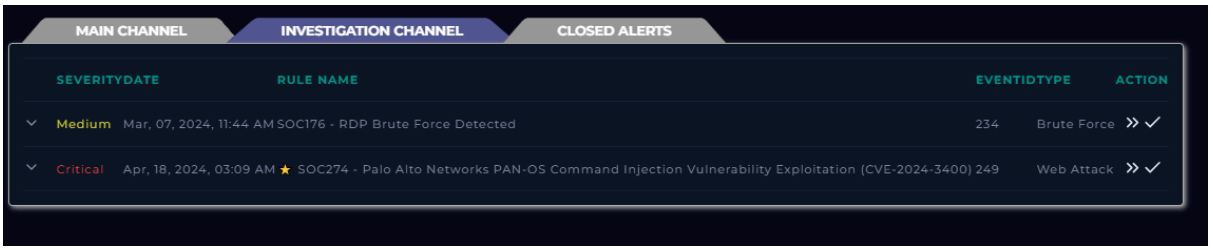
Remote Desktop Protocol (RDP) Brute Force Investigation Report

1.0 Incident Summary



On March 7, 2024, a medium-severity SIEM alert identified suspicious Remote Desktop Protocol (RDP) authentication activity targeting a Windows 10 client system. The alert was triggered after multiple failed login attempts originating from a single external IP address against several non-existent user accounts.

Initial investigation confirmed the activity as an RDP brute force attack. Log analysis revealed multiple failed authentication attempts followed by one successful login, indicating that the attacker successfully gained access to the endpoint.

Further endpoint analysis identified suspicious process manipulation activity involving critical Windows authentication-related processes, including logonui.exe, usosvc, and winlogon.exe. Command-line history and network activity suggested post-compromise reconnaissance and possible credential harvesting attempts.

Due to evidence of successful compromise and post-authentication activity, the incident was escalated to the Incident Response (IR) team for deeper forensic investigation and containment.

2.0 Alert Details

Field	Value
Alert Name	Criticial RDP Brute Force Alert
Severity	Medium
Detection Source	SIEM
Event ID	234
Date/Time	March 7, 2024 - 11:44am
Source IP	218.92.0.56
Destination Host	MATTHEW
Operating System	Windows 10 (64-bit)
Primary User	Matthew
Trigger Condition	Login failure from a single source with different non-existing accounts

3.0 Initial Triage

The investigation began after the SIEM generated an alert for repeated RDP login failures originating from a single source IP address. The pattern of authentication attempts targeting multiple usernames strongly suggested brute force activity.

Review of authentication logs revealed:

- 14 failed login attempts
- 1 successful authentication attempt
- Multiple login attempts against non-existent accounts
- Repeated use of common dictionary-based usernames including:
    - guest
    - admin
    - Matthew
    - sysadmin

The successful authentication event indicated that the attacker was ultimately able to gain remote access to the endpoint.

4.0 Observed Activity

4.0.1 Authentication Activity

Observed authentication behavior aligned with typical brute force techniques:

- Multiple rapid failed authentication attempts

- Repeated username guessing activity
- Successful login after multiple failures

4.0.2 Process Activity

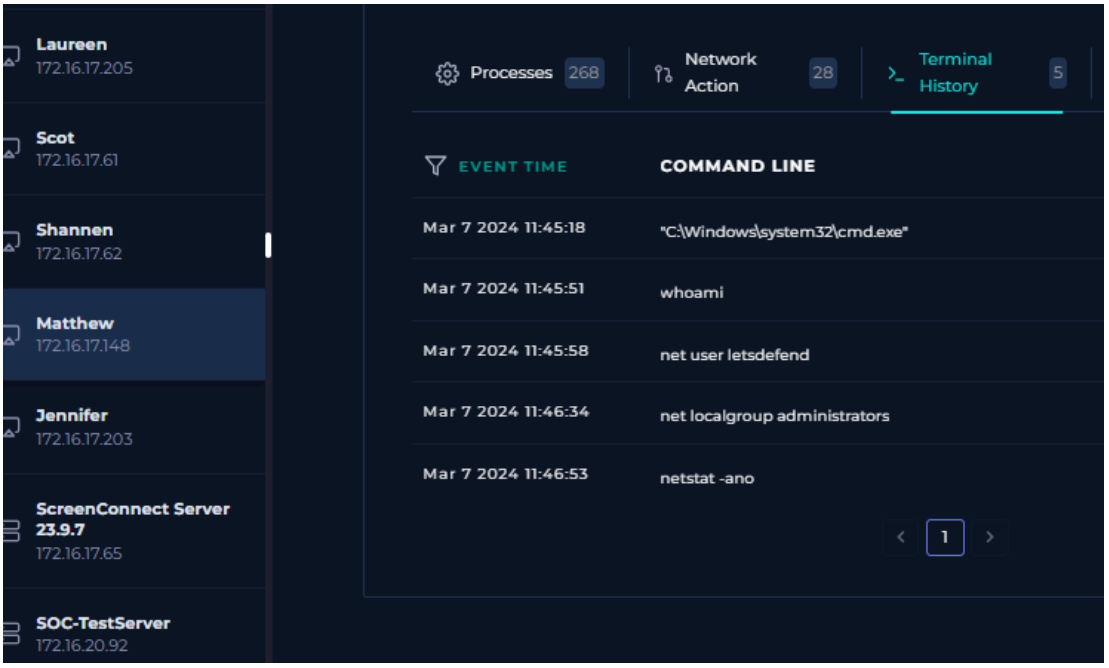
Endpoint analysis identified suspicious interaction with the following processes:

- logonui.exe
- usosvc
- winlogon.exe

The manipulation of authentication and login-related processes suggested possible:

- credential access attempts,
- persistence activity,
- or system-level tampering following successful authentication.

4.0.3 Command-Line Activity



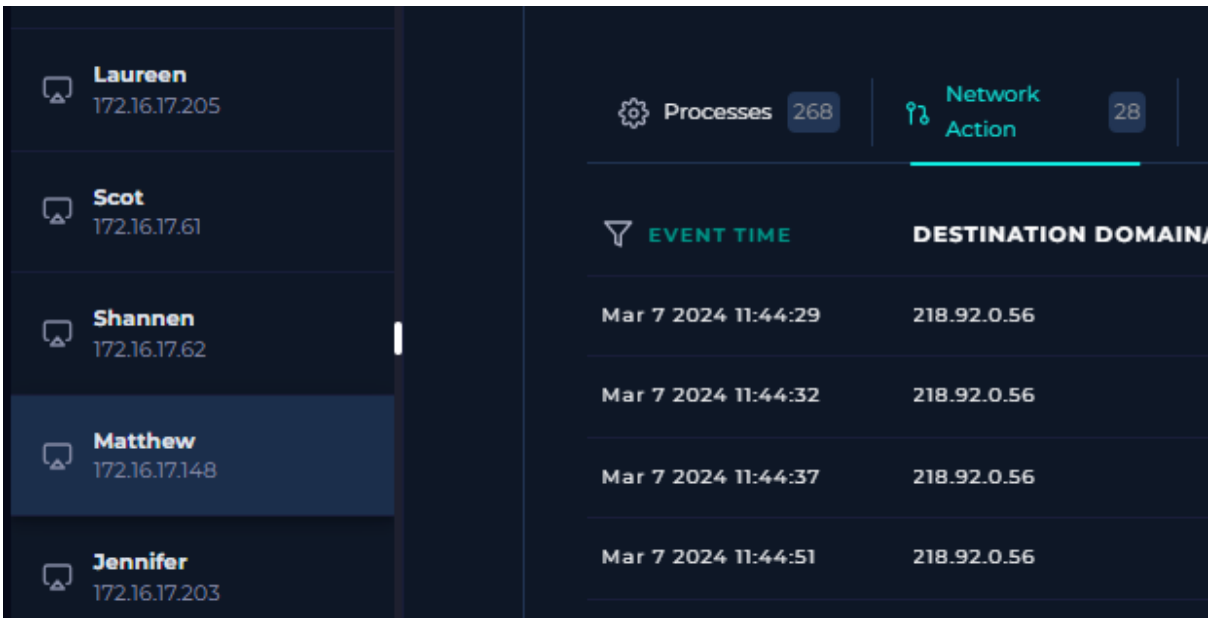
Terminal history identified several reconnaissance commands executed on the compromised host:

- whoami
- net user letsdefend
- net localgroup administrators
- netstat -ano

These commands are commonly associated with:

- user enumeration,
- privilege discovery,
- administrative group identification,
- and network reconnaissance.

4.0.4 Network Activity



Network analysis identified multiple outbound connections initiated from the host following successful authentication.

Observed behavior suggested:

- active communication with external infrastructure,
- possible command-and-control activity,

- or remote attacker interaction with the compromised endpoint.

Further traffic inspection was recommended during escalation to the Incident Response team.

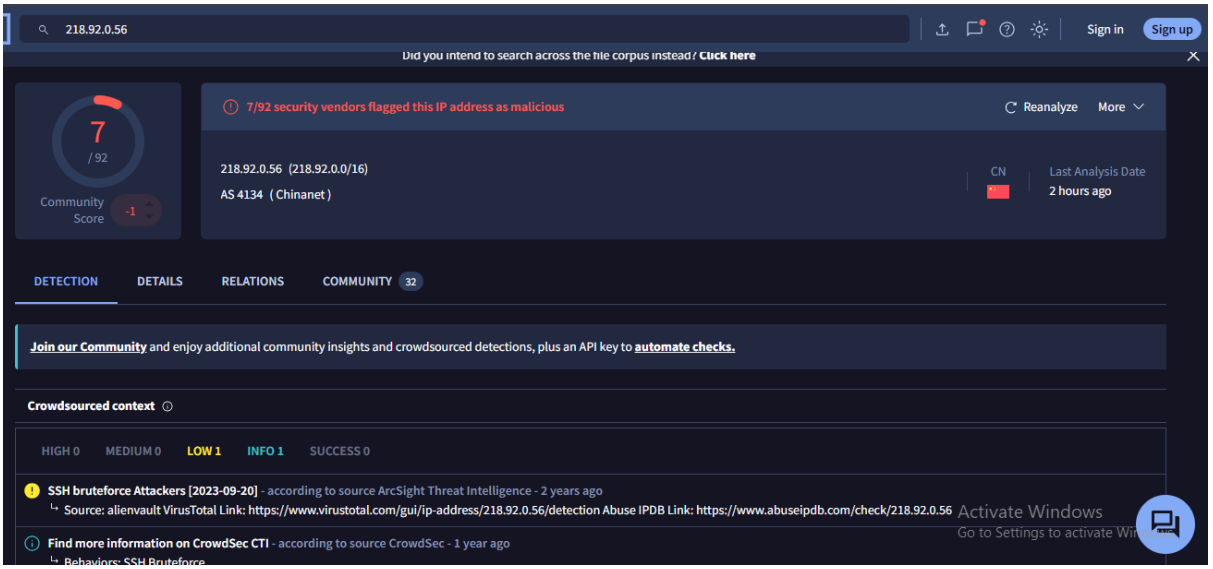
5.0 IOC Extraction

IOC Type	Value
Source IP Address	218.92.0.56
Destination IP Address	172.16.171.148
Destination Host	Matthew
Attack Method	RDP Brute Force
Processes Observed	logonui.exe, usosvc, winlogon.exe
Commands Observed	whoami, net user, net localgroup administrators, netstat -ano

6.0 Threat Intelligence Enrichment

Threat intelligence enrichment was performed on the identified external IP address to determine its reputation and potential association with malicious activity.

6.0.1 Source IP Analysis - 218.92.0.56

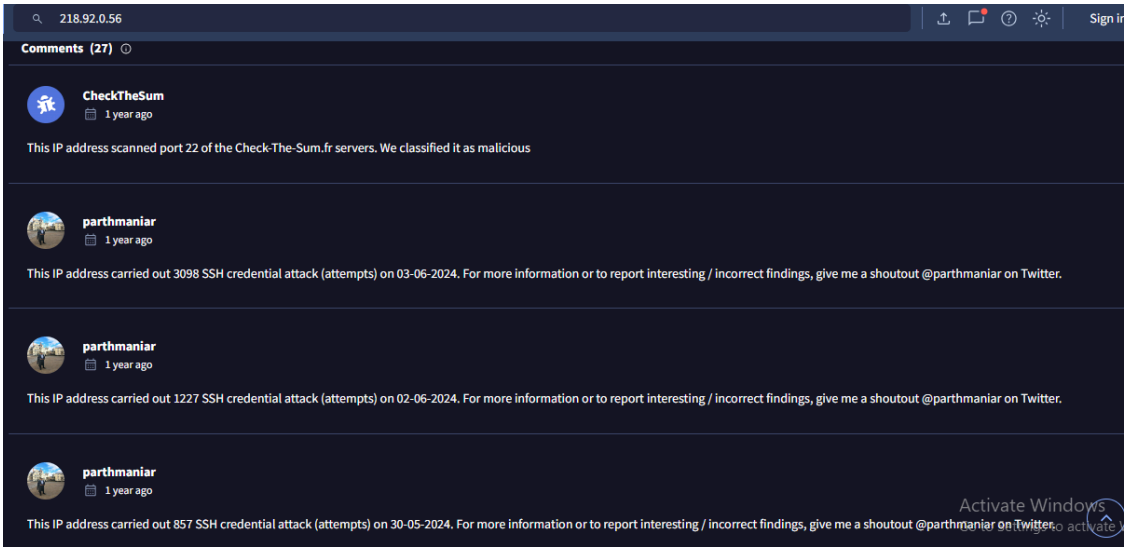


Threat intelligence findings indicated:

- Historical association with brute force scanning activity
- Multiple reports of unauthorized authentication attempts
- Previous involvement in suspicious RDP-related activity
- Reputation indicators consistent with hostile external scanning infrastructure

The activity pattern observed during the investigation aligned with known brute force attack behavior commonly used to gain unauthorized remote access.

6.0.2 Behavioral Correlation



The observed attacker activity aligned with common post-compromise techniques, including:

- account enumeration,
- privilege discovery,
- system reconnaissance,
- and credential-related activity.

The combination of successful RDP access and post-authentication reconnaissance significantly increased confidence that the endpoint had been compromised.

7.0 Timeline of Events

Time	
11:44 am	SIEM alert triggered for repeated RDP login failures
11:45 am	Multiple failed authentication attempts identified
11:54 am	Successful RDP login observed
12:02 am	Suspicious process activity identified
12:07 am	Command-line reconnaissance activity observed
12:08 am	Outbound network connections detected
12:10 am	Host assessed as potentially compromised
12:15 am	Incident escalated to Incident Response team

8.0 MITRE ATT&CK Mapping

Technique ID	Technique
T1110	Brute Force
T1021.001	Remote Services: Remote Desktop Protocol
T1087	Account Discovery
T1069.001	Permission Groups Discovery: Local Groups
T1016	System Network Configuration Discovery
T1033	System Owner/User Discovery
T1057	Process Discovery

9.0 Analysis & Findings

The investigation confirmed that the detected activity represented a successful RDP brute force compromise against the Windows 10 endpoint MATTHEW.

Several factors contributed to this assessment:

- Multiple failed login attempts from a single external source
- Username guessing activity using common account names
- Successful authentication following repeated failures
- Post-authentication reconnaissance commands
- Suspicious manipulation of Windows authentication-related processes
- External network communication initiated after compromise

The attacker demonstrated behavior consistent with early-stage post-compromise reconnaissance and environment discovery.

Although evidence of lateral movement or malware deployment was not identified during the scope of this investigation, the successful remote access and subsequent activity indicated that the endpoint should be treated as compromised.

Due to the possibility of credential exposure and further attacker persistence, the target host was contained according to letsefend policy and escalation to the Incident Response team was determined necessary for deeper forensic analysis.

10.0 Recommendations

The following actions were recommended:

1. Disable or restrict exposed RDP access
2. Reset credentials associated with the affected host
3. Block the identified external IP address at perimeter controls
4. Conduct full forensic analysis of the compromised endpoint
5. Enable multi-factor authentication (MFA) for remote access services
6. Implement account lockout policies to reduce brute force exposure